

ABSTRACT

The Dark Web hosts a wide range of illicit activities, making efficient and accurate text classification essential for monitoring and threat intelligence. Existing approaches that combine TextCNN with Labeled Latent Dirichlet Allocation (LLDA) rely heavily on class-dependent topic modeling and keyword-based embedding construction. Although effective, such methods suffer from restricted vocabulary coverage, loss of semantic information, and limited adaptability to emerging Dark Web terminology. To address these limitations, this work proposes a hybrid Dark Web classification framework integrating TF-IDF, unsupervised LDA, GloVe embeddings, and TextCNN. TF-IDF is applied to emphasize informative terms, while LDA captures latent thematic structures independent of class labels. Pre-trained GloVe embeddings provide rich semantic representations, and TextCNN extracts discriminative local features from weighted embedding sequences. The LDA document-topic vectors are fused with the CNN feature representations to incorporate both global and local contextual cues. Experimental analysis shows that this multi-feature fusion approach enhances robustness, improves semantic understanding, and achieves superior classification accuracy compared to the existing LLDA-guided TextCNN model.

Keywords: Dark Web Classification, TF-IDF, LDA, GloVe Embeddings, TextCNN, Topic Modeling, Deep Learning, Illicit Content Detection.

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